

How Well Did We Do?

Measuring Digital Preservation Performance

Andrea Goethals · Martin Gengenbach · Cynthia Wu
National Library of New Zealand & Archives New Zealand (ALNZ)

A practical answer to a difficult question

Digital preservation programmes often report how much content has been ingested or seek to demonstrate repository ‘trustworthiness’. These approaches say little about whether digital material is actually being maintained in accordance with the institution’s preservation policy. We developed a performance metric that combines two factors: the proportion of digital holdings stored in compliant preservation environments, and evidence that those environments were actively managed and maintained during the reporting period. Together, they provide a relatively simple way to evidence and communicate digital preservation performance across different collections.

Why we needed this measure

At ALNZ, a new organisational measure was proposed: ‘Percentage of collections and holdings that are maintained according to the policies for collections and holdings care.’ This required comparable approaches for physical and digital collections.

For physical collections, preservation conditions are commonly evaluated through environmental monitoring such as temperature and humidity. The digital equivalent was less obvious. We needed to assess both whether collections were held in appropriate preservation environments and whether those environments continued to operate as intended.

The conceptual model

For digital collections, determining whether holdings are maintained in accordance with preservation policy combine two considerations:

- 1

Percentage in compliant preservation environments
Is the content held in an environment that meets our institution’s mandatory preservation requirements defined for that collection?
- 2

Operational assurance
Is there evidence that the environment was actively managed and maintained as intended during the reporting period?

How the metric works

1 - Percentage in compliant preservation environments

This component measures the percentage of digital holdings stored in environments that meet the mandatory preservation requirements defined for the collection. These requirements were developed through a review of established standards and shared best practices, such as the Digital Preservation Storage Criteria, as well as consultation with collection stakeholders.

At ALNZ, these requirements relate to:

Preservation storage	Object integrity
Preservation metadata	Access control

The requirements may be defined differently for different types of collections.

Other institutions should select requirements appropriate to their policies and operating context. The important point is to define what compliance means before measuring coverage.

2 - Operational assurance

This component assesses whether each mandatory preservation requirement was actively managed and maintained during the reporting period. Evidence is considered against three questions:

Ownership	Is responsibility for overseeing the requirement and addressing issues clearly assigned?
Monitoring	Is compliance actively monitored through defined processes capable of identifying issues requiring attention?
Response	Is there a defined and operating process for investigating, escalating, recording and remediating issues?

Evidence may consist of either design evidence - relatively stable documentation such as job descriptions, policies and procedures - or operational evidence produced during the reporting period, such as logs, dashboard screenshots and emails.

Together, these questions test whether preservation requirements are supported by effective operational arrangements—not merely documented or technically configured.

Final digital preservation performance score

Percentage in compliant preservation environments × Operational assurance score

Both components are necessary: technical compliance is insufficient without effective management, while strong operations cannot compensate for content held outside compliant preservation environments.

Worked example: the same framework, two collections

Each collection can define its requirements differently and be assessed separately; the results are then combined using a volume-weighted average.

Preservation environment compliance

The **Digital Repository** (ALNZ’s cultural heritage collections and government records) and **Whole of Domain** web archive use the same four requirement areas, but some are defined differently. Integrity information, metadata and access restrictions are recorded and checked differently in Rosetta/METS and in WARC-based workflows.

All content in both collections was assessed as being in compliant preservation environments: **100% coverage**.

Operational assurance

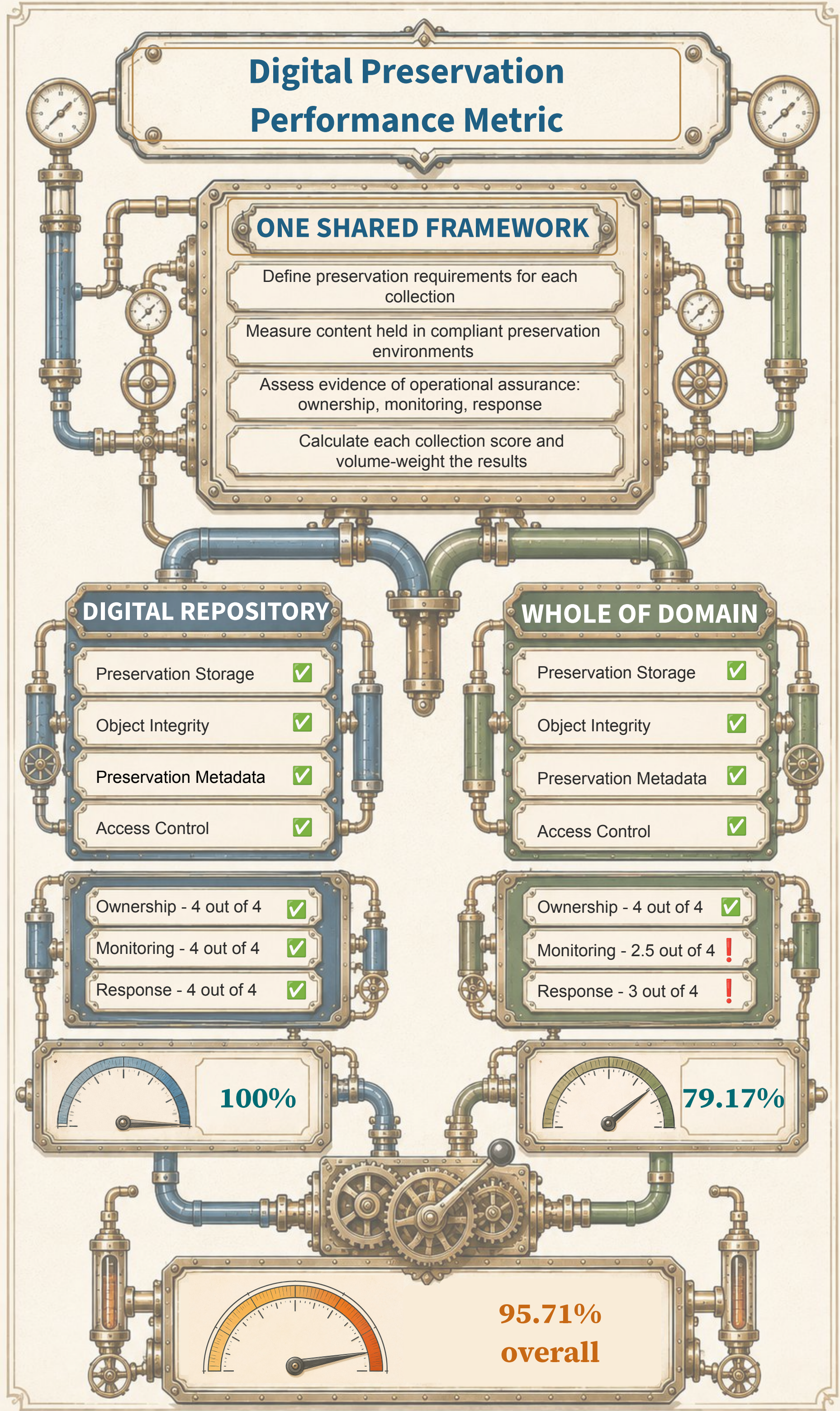
Evidence for each of the four requirements was assessed against Ownership, Monitoring and Response. A ‘yes’ earned 1 point, ‘in part’ 0.5 points and ‘no’ 0 points.

Digital Repository: 12/12 = **100%**
Whole of Domain web archive: 9.5/12 = **79.17%**

Volume-weighted result

Digital Repository: 1,357 TB / **79%** of preserved volume
Whole of Domain: 352 TB / **21%** of preserved volume

$$(100\% \times 79\%) + (79.17\% \times 21\%) = 95.71\%$$



Why this approach may be useful elsewhere

The metric provides a single indicator while allowing requirements to be adapted to different collections. It can be applied to digitised and born-digital material, digital records and large-scale web archives - and adapted by other institutions to fit their own policies and operating context.

Interested in trying it with your collections? Contact us!

Andrea Goethals · andrea.goethals@dia.govt.nz
Cynthia Wu · cynthia.wu@dia.govt.nz
Martin Gengenbach · martin.gengenbach@dia.govt.nz